RESEARCH PROJECT PROPOSAL

Explainable AI for Investment Decision-Making in the Canadian Equity Market

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Table of Contents

1. Introduction 3

2. Literature Review 4

3. Data Set and Variable Definitions 5

3.1 Unit of analysis and timing 5

4. Methodology 6

4.1 Prediction targets and feature engineering 6

4.2 Data preparation and controls 6

4.3 Model development, validation, and evaluation 7

4.4 Explainable AI and research outputs 7

5. Project Plan and Gantt Chart 9

Explainable AI (XAI) for Investment Decision-Making in the Canadian Equity Market

1. Introduction

Artificial intelligence (AI) is becoming increasingly useful in financial markets, helping investors analyze large amounts of information and identify patterns that traditional methods might miss. However, there is one major problem: AI models can make predictions without clearly explaining how they reached them. This is commonly known as the *black-box problem*.

This project, titled "Explainable AI (XAI) for Investment Decision-Making in the Canadian Equity Market," aims to investigate whether machine learning can predict stock performance and investment risk while making those predictions easier to understand.

The study will focus on selected companies from the S&P/TSX Composite, Canada's main stock market benchmark. Two outcomes will be examined. First, the model will predict whether a stock will outperform the S&P/TSX Composite in the following month. This is measured using *active return*, which is the difference between a stock's return and the market's return. Second, the model will predict whether a stock will experience high volatility over the following 12 months. *Volatility* measures how much stock returns fluctuate, and annualized volatility of 20% or greater will be used as the high-risk threshold, adapted from Canadian investment fund risk classifications.

To make these predictions, the study will combine company fundamentals, such as return on assets (ROA), return on equity (ROE), earnings growth, and price-to-earnings (P/E) ratios, with monthly stock returns and macroeconomic indicators such as inflation and interest rates.

Two models will be compared: logistic regression, a traditional statistical method for predicting the probability of an outcome, and XGBoost, a more advanced machine-learning algorithm that combines decision trees to identify complex relationships. Their performance will be tested using historical data that was not used during training, known as *out-of-sample testing*.

The main focus, however, is explainability. SHAP and LIME, two Explainable AI (XAI) techniques, will help identify which financial and market variables influenced XGBoost's predictions. The study will also investigate whether these factors differ across sectors, since banks, technology companies, and utilities operate under different financial conditions.

Ultimately, the goal is not simply to identify which stocks might perform well, but to determine how accurately AI can predict investment outcomes and how clearly we can explain its reasoning. The findings may help make machine-learning-based investment analysis more transparent and useful for financial decision-making.

2. Literature Review

Predicting stock performance has been an important topic in financial research for decades. Traditional financial models often examine how company fundamentals, valuation, and market conditions relate to stock returns. Fama and French (1992) found that company size and book-to-market ratio, which compares a company's accounting value with its market value, help explain differences in stock returns. They also recognized that financial ratios can have different meanings across industries. For example, high leverage is normal for banks but may indicate financial difficulty for other companies. This is particularly relevant to our study, which includes companies from different Canadian sectors.

More recently, researchers have explored whether machine learning can improve financial predictions. Gu et al. (2020) examined several machine-learning methods for predicting monthly stock returns using 94 company characteristics, including financial ratios, valuation measures, and market indicators. Their study demonstrated how annual, quarterly, and monthly information can be combined in one dataset. They also accounted for reporting delays to avoid *look-ahead bias*, which occurs when a model accidentally uses information that was not available at the time of prediction. This provides an important foundation for our project, since company financial statements are reported annually or quarterly, while our stock-return data is monthly.

One promising machine-learning technique is XGBoost, introduced by Chen and Guestrin (2016). XGBoost uses multiple decision trees, with each new tree attempting to improve the predictions of earlier trees. Unlike simpler statistical models, it can capture *nonlinear relationships*, meaning that changes in one financial variable do not necessarily have the same effect under every market condition. Although this flexibility can improve prediction, it can also make the model difficult to interpret.

Babaei and Giudici (2025) addressed this challenge in their study of Explainable AI for investment decision-making. They applied XGBoost to financial data from 2,049 small and medium-sized enterprises to estimate default risk and examine expected profitability. They then used Shapley values to explain which financial indicators contributed to the model's predictions. Their research demonstrates how AI can support financial analysis while providing explanations instead of only predictions. Our project builds on this approach but applies it to publicly traded Canadian stocks, focusing on market outperformance and volatility-based risk rather than company default.

To better understand black-box predictions, researchers have developed Explainable AI (XAI) techniques. Lundberg and Lee (2017) introduced SHAP (SHapley Additive exPlanations), which measures how individual variables contribute to a prediction. For example, SHAP can help explain whether a company's profitability, recent returns, or valuation increased or decreased its predicted probability of outperforming the market.

Similarly, Ribeiro et al. (2016) developed LIME (Local Interpretable Model-agnostic Explanations), which explains individual predictions by approximating the complex model with a simpler one around a particular observation. While SHAP can provide both overall and company-specific explanations, LIME is mainly used to explain individual predictions.

Together, these studies provide the foundation for our research. However, accurate predictions do not necessarily mean that a model's decisions are understandable or economically meaningful. Therefore, this project will compare logistic regression and XGBoost in predicting Canadian stock outperformance and risk, then apply SHAP and LIME to examine the factors behind those predictions. By comparing explanations across sectors, the study aims to determine whether Explainable AI can make complex financial predictions more transparent and useful for investment decision-making.

3. Data Set and Variable Definitions

Population and sampling. The initial collection list contains 110 candidate TSX companies across 11 sectors. The usable sample may fall to roughly 60–110 firms after Bloomberg coverage, listing history, missingness, and data-quality checks. The target study period is January 2013–December 2024 (144 monthly observations per continuously observed stock), or at most 15,840 company-month rows for 110 firms and 8,640 for 60. These are upper bounds, not an assertion that every security has a complete history. Companies entering, leaving, restructuring, or delisting during the period will be documented to limit survivorship bias.

Source and frequency	Variables (examples)	Treatment in company-month dataset
Bloomberg: annual/quarterly	ROA, ROE, profit margin, EBIT, revenue, revenue growth, net income growth, EPS growth, dividend payout, debt ratios	Use latest information after release; retain fiscal year and publication/eligibility date.
Bloomberg: monthly market	Total-return series, shares/volume, P/E, P/B, market capitalization, dividend yield	Update at each month-end; historical P/E must use trailing earnings known then.
Bloomberg: derived market	Past 1/3/6/12-month returns, active returns, rolling beta, volatility, active risk, drawdown, turnover	Calculate with completed historical months only.
Market and macro	S&P/TSX Composite total return, sector returns, rates, CPI, GDP, oil, gold, CAD/USD	Use release-date-aligned values, with published information carried forward.
Constructed outcomes	Next-month TSX outperformance; forward 12-month high volatility	Created from future returns solely as response labels, never contemporaneous inputs.

Bloomberg financial fields being collected include RETURN_ON_ASSET, RETURN_COM_EQY, PROF_MARGIN, EBIT_AFTER_OPERATING_LEASE, SALES_REV_TURN, SALES_GROWTH, NET_INCOME_TO_COMMON_SEQ_GROWTH, EPS_DIL_BEF_EXTRAORD_SEQ_GRWTH, DVD_PAYOUT_RATIO, TOT_DEBT_TO_TOT_EQY, TOT_DEBT_TO_TOT_ASSET, DIVIDEND_12_MONTH_YIELD, HISTORICAL_MARKET_CAP, and PX_TO_BOOK_RATIO. These are raw collection definitions, not a final promise that every feature is economically comparable or available for every sector. For example, Bloomberg defines profit margin differently for banks, and conventional leverage and EBIT ratios are not directly comparable between banks and industrial firms.

3.1 Unit of analysis and timing

Each row represents stock i at the end of month t. Stock and benchmark returns are measured monthly and should both include dividends when constructing benchmark-relative performance. Annual or quarterly accounting ratios retain the most recent published value until a new disclosure is available. If original release dates cannot be recovered, a documented conservative delay will be used (e.g., six months for annual and four months for quarterly characteristics, following Gu et al., 2020). Macroeconomic values will similarly be matched by their actual release dates, not their reference periods.

For example, RBC at January 2020 month-end is paired with its latest published financial ratios and historical returns through January; the outcome is whether RBC beats the TSX in February 2020. This illustrates information availability rather than actual RBC values.

4. Methodology

4.1 Prediction targets and feature engineering

Two different binary response variables will be estimated using separate models. The first asks whether an individual stock delivers a larger total return than the benchmark next month. The second asks whether that stock experiences high total return volatility during the following 12 months. These are distinct tasks: a stock may outperform the benchmark while still being highly volatile.

$$\text{Active return}_{i,t+1} = R_{i,t+1} - R_{m,t+1}$$ *R_m = S&P/TSX Composite total return*

$$\text{Outperform}_{i,t+1} = 1 \text{ if Active return}_{i,t+1} > 0; \text{ otherwise } 0$$

$$\text{Future volatility}_{i,t} = SD(R_{i,t+1} \ldots R_{i,t+12}) \times \sqrt{12}$$

$$\text{HighRisk}_{i,t} = 1 \text{ if Future volatility}_{i,t} \geq 0.20; \text{ otherwise } 0$$

Monthly returns will be based on consistent adjusted/total-return information to account for splits and dividends. Predictors will include past cumulative returns (momentum), past active returns, historical volatility, beta relative to the TSX, active risk (annualized standard deviation of benchmark-relative returns), and maximum drawdown. Rolling risk features will use observations ending at month t; 12-month and longer lookback periods will be compared where appropriate. A 36- or 60-month beta is an estimation choice for that feature, not a restriction on the 2013–2024 study period.

The 20% high-risk cutoff is an adaptation of Appendix F of National Instrument 81-102, which labels mutual funds with historical standard deviation of 20% or more "High." It is not an official classification of individual stocks and is calculated over a different horizon here. Therefore, we will report a sensitivity check at 25% and describe the 20% cutoff as a research definition rather than a regulatory rule for equities (Ontario Securities Commission, 2024).

4.2 Data preparation and controls

First, Bloomberg data will be checked for missing values, duplicate ticker-month observations, inconsistent units (percent versus decimals; CAD versus USD), corporate actions, changing tickers, and extreme ratios caused by near-zero or negative earnings. No missing financial value will be replaced with zero. We will examine missingness by stock, year, and feature; unavailable bank-specific fields will not automatically cause bank exclusions. A smaller unbalanced sample is acceptable when its historical coverage is documented.

Second, predictors will be aligned to the information set available at each month-end. Historical P/E and P/B are month-end valuations; annual financial ratios will not be assigned backward to months before their publication. Repeated annual figures across several months are expected and will not be treated as separate financial disclosures.

Third, variables will be selected using financial meaning and training-only data diagnostics. High-missingness fields, unstable ratios, and highly correlated measures will be reviewed; variance inflation factor (VIF) may be used to diagnose multicollinearity for logistic regression. The XGBoost model will also include sector indicators. Because the same variables may matter differently for banks, energy, technology, and utilities, common factors will be interpreted both globally and within each sector, while sector-specific measures such as CET1 will remain outside the principal all-sector model.

4.3 Model development, validation, and evaluation

Baseline and main model. Logistic regression provides a transparent benchmark for predicting the probability of each yes/no outcome. XGBoost (eXtreme Gradient Boosting) is the primary algorithm: it sequentially builds decision trees to improve predictions and uses regularization to control complexity (Chen & Guestrin, 2016). Key hyperparameters include maximum tree depth, number of trees, learning rate, subsampling, and minimum child weight. Tuning will be based on validation results only.

Temporal separation. Predictions must be assessed on future periods not used to estimate or tune the model. The broad chronology will be training years 2013–2020, validation 2021–2022, and final test 2023–2024. Exact eligible months differ by target: for next-month outperformance, a November 2024 predictor row may be labelled by December 2024. For the 12-month risk target, a December 2024 predictor row would require data through December 2025 and therefore cannot be evaluated using a dataset ending in 2024. To prevent label overlap, the risk model will train using feature dates through December 2019 (future-risk labels end by December 2020), validate on 2021 origin dates (labels finish in 2022), and finally test on 2023 origin dates (labels finish in 2024). These holdout partitions will be fixed before analysis.

Metric	Definition	Role in our two classification models
ROC-AUC	Area under the receiver operating characteristic curve; ranks positives above negatives across probability cutoffs.	Primary discrimination measure for logistic regression vs XGBoost.
Precision = TP/(TP + FP)	Share of positive predictions that were truly positive.	Avoid too many false outperformance or high-risk signals.
Recall = TP/(TP + FN)	Share of true positives that the model identifies.	Important for not overlooking stocks that turn out high risk.
Confusion matrix	Counts TP, TN, FP and FN after choosing a probability threshold.	Shows false alarms and missed events; threshold tuned on validation.
PR-AUC / balanced accuracy	Helpful where positive and negative classes are unequal.	Secondary checks if class imbalance is substantial.

Model comparison will report validation and final test ROC-AUC, precision, recall, and confusion matrices separately for the two outcomes, with class prevalence. Any reweighting or imputation will be learned from training data only. We will test class weights before considering SMOTE, because synthetic observations across time and company identities require special care. The final test will not be used to repeatedly revise hyperparameters or feature definitions.

4.4 Explainable AI and research outputs

SHAP will quantify how each predictor shifts a model prediction relative to a baseline. We will report global mean absolute SHAP contributions, separate SHAP summaries for each sector with adequate observations, and local waterfall plots for selected stock-month examples. Sector-balanced summaries will be considered so a sector with many firms does not automatically dominate the pooled importance ranking. LIME will approximate the XGBoost prediction locally for the same selected examples, allowing a qualitative comparison of important explanatory variables (Lundberg & Lee, 2017; Ribeiro et al., 2016). Alternatives include annual-only modelling (less temporal

detail) and MIDAS/deep learning (more complexity). Limitations include survivorship bias, sector imbalance, correlated months, and adapting a fund-risk cutoff; SHAP does not establish causality.

5. Project Plan and Gantt Chart

Proposed nine-week schedule beginning 12 October 2026. The instructor's submission date has not been specified in the supplied requirements, so week numbers and dates are provisional. Workstreams overlap deliberately: reading and writing proceed alongside Bloomberg extraction and model construction. Milestones are a validated dataset by W5, a selected model by W7, and final written report and poster presentation by W9.

Work package	W1 Oct 12	W2 Oct 19	W3 Oct 26	W4 Nov 2	W5 Nov 9	W6 Nov 16	W7 Nov 23	W8 Nov 30	W9 Dec 7
1. Research question, scope and reviewed sources	■	■							
2. Bloomberg extraction + ticker/sector checks	■	■	■						
3. Audit N/A, reporting dates and sample attrition		■	■	■					
4. Construct monthly panel and publication lags			■	■	■				
5. Total returns, active returns, beta/risk features			■	■	■				
6. Next-month and next-12M target definitions				■	■				
7. Train-only cleaning + feature diagnostics					■	■			
8. Logistic baseline and XGBoost tuning						■	■		
9. Final out-of-sample testing + sensitivity checks							■	■	
10. SHAP/LIME, sector analysis and results figures							■	■	
11. Writing, APA citations and revision	■	■	■		■	■	■	■	■
12. Poster outline, key results and visual design							■	■	
13. Poster feedback, revisions and presentation rehearsal								■	■
14. Final report submission and poster presentation									■

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